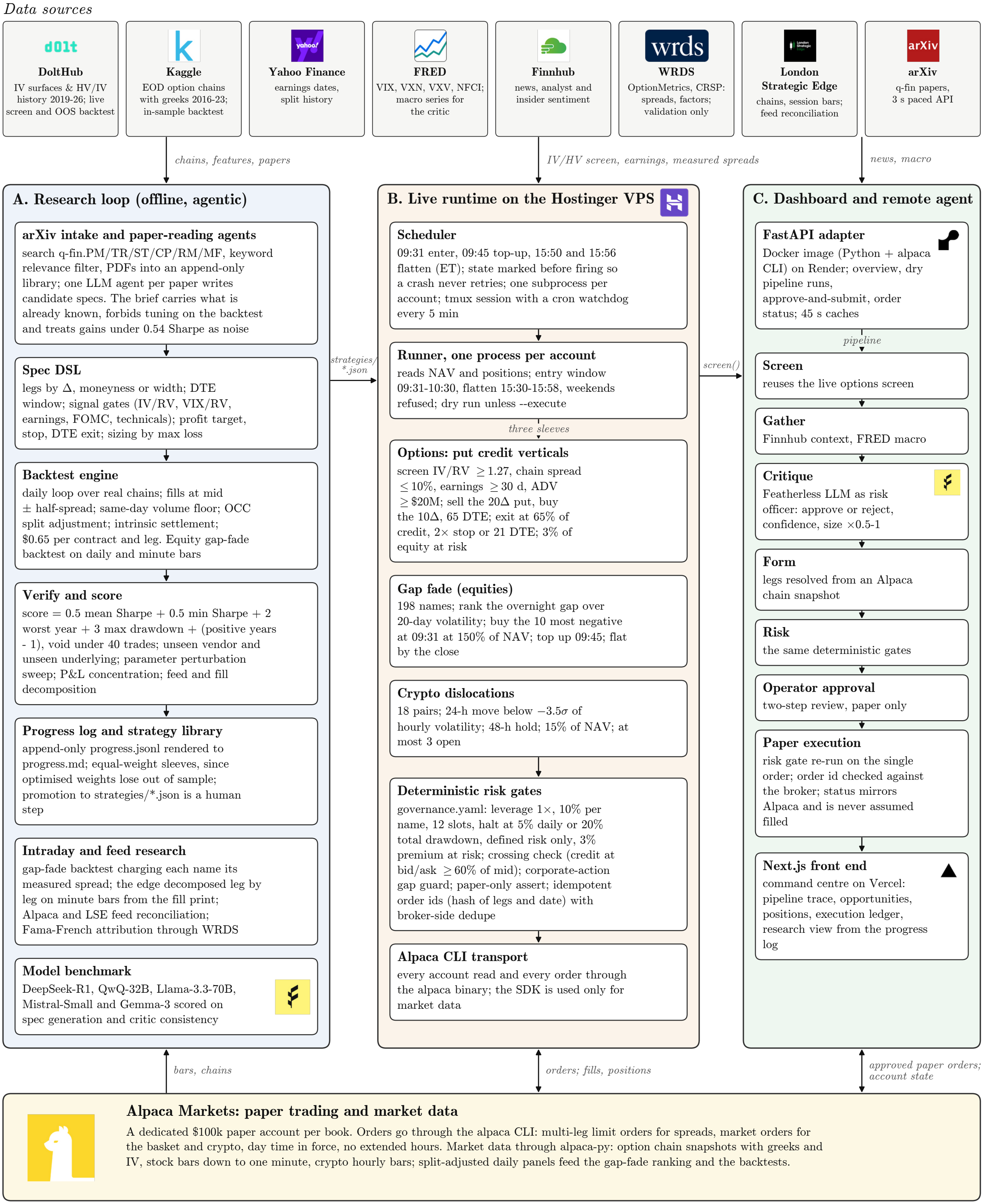

Data sources
DoltHub
IV surfaces & HV/IV history 2019-26; live screen and OOS backtest
Kaggle
EOD option chains with greeks 2016-23; in-sample backtest
Yahoo Finance
earnings dates, split history
FRED
VIX, VXN, VXV, NFCI; macro series for the critic
Finnhub
news, analyst and insider sentiment
WRDS
OptionMetrics, CRSP: spreads, factors; validation only
London Strategic Edge
chains, session bars; feed reconciliation
arXiv
q-fin papers, 3 s paced API
chains, features, papers
IV/HV screen, earnings, measured spreads
news, macro
A. Research loop (offline, agentic)
arXiv intake and paper-reading agents
search q-fin.PM/TR/ST/CP/RM/MF, keyword relevance filter, PDFs into an append-only library; one LLM agent per paper writes candidate specs. The brief carries what is already known, forbids tuning on the backtest and treats gains under 0.54 Sharpe as noise
Spec DSL
legs by Δ, moneyness or width; DTE window; signal gates (IV/RV, VIX/RV, earnings, FOMC, technicals); profit target, stop, DTE exit; sizing by max loss
Backtest engine
daily loop over real chains; fills at mid ± half-spread; same-day volume floor; OCC split adjustment; intrinsic settlement; $0.65 per contract and leg. Equity gap-fade backtest on daily and minute bars
Verify and score
score = 0.5 mean Sharpe + 0.5 min Sharpe + 2 worst year + 3 max drawdown + (positive years - 1), void under 40 trades; unseen vendor and unseen underlying; parameter perturbation sweep; P&L concentration; feed and fill decomposition
Progress log and strategy library
append-only progress.jsonl rendered to progress.md; equal-weight sleeves, since optimised weights lose out of sample; promotion to strategies/*.json is a human step
Intraday and feed research
gap-fade backtest charging each name its measured spread; the edge decomposed leg by leg on minute bars from the fill print; Alpaca and LSE feed reconciliation; Fama-French attribution through WRDS
Model benchmark
DeepSeek-R1, QwQ-32B, Llama-3.3-70B, Mistral-Small and Gemma-3 scored on spec generation and critic consistency
strategies/ *.json
B. Live runtime on the Hostinger VPS
Scheduler
09:31 enter, 09:45 top-up, 15:50 and 15:56 flatten (ET); state marked before firing so a crash never retries; one subprocess per account; tmux session with a cron watchdog every 5 min
Runner, one process per account
reads NAV and positions; entry window 09:31-10:30, flatten 15:30-15:58, weekends refused; dry run unless --execute
three sleeves
Options: put credit verticals
screen IV/RV ≥ 1.27, chain spread ≤ 10%, earnings ≥ 30 d, ADV ≥ $20M; sell the 20Δ put, buy the 10Δ, 65 DTE; exit at 65% of credit, 2× stop or 21 DTE; 3% of equity at risk
Gap fade (equities)
198 names; rank the overnight gap over 20-day volatility; buy the 10 most negative at 09:31 at 150% of NAV; top up 09:45; flat by the close
Crypto dislocations
18 pairs; 24-h move below −3.5σ of hourly volatility; 48-h hold; 15% of NAV; at most 3 open
Deterministic risk gates
governance.yaml: leverage 1×, 10% per name, 12 slots, halt at 5% daily or 20% total drawdown, defined risk only, 3% premium at risk; crossing check (credit at bid/ask ≥ 60% of mid); corporate-action gap guard; paper-only assert; idempotent order ids (hash of legs and date) with broker-side dedupe
Alpaca CLI transport
every account read and every order through the alpaca binary; the SDK is used only for market data
screen()
C. Dashboard and remote agent
FastAPI adapter
Docker image (Python + alpaca CLI) on Render; overview, dry pipeline runs, approve-and-submit, order status; 45 s caches
pipeline
Screen
reuses the live options screen
Gather
Finnhub context, FRED macro
Critique
Featherless LLM as risk officer: approve or reject, confidence, size ×0.5-1
Form
legs resolved from an Alpaca chain snapshot
Risk
the same deterministic gates
Operator approval
two-step review, paper only
Paper execution
risk gate re-run on the single order; order id checked against the broker; status mirrors Alpaca and is never assumed filled
Next.js front end
command centre on Vercel: pipeline trace, opportunities, positions, execution ledger, research view from the progress log
bars, chains
orders; fills, positions
approved paper orders; account state
Alpaca Markets: paper trading and market data
A dedicated $100k paper account per book. Orders go through the alpaca CLI: multi-leg limit orders for spreads, market orders for the basket and crypto, day time in force, no extended hours. Market data through alpaca-py: option chain snapshots with greeks and IV, stock bars down to one minute, crypto hourly bars; split-adjusted daily panels feed the gap-fade ranking and the backtests.